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UNIVERSITY OF TORONTO
Faculty of Arts and Science

APRIL 2999 EXAMINATIONS

CSC336H1S

Duration — 3 hours

Examination Aids: **One $8\frac{1}{2}'' \times 11''$ sheet of paper, handwritten on both sides.**
No photocopies.
A calculator.

Student Number: _____

Last (Family) Name(s): _____

First (Given) Name(s): _____

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*Do **not** turn this page until you have received the signal to start.*
(In the meantime, please fill out the identification section above, and read the instructions below *carefully*.)

This final examination consists of 9 questions on 14 pages (including this one). When you are told to start, please make sure that your copy of the examination is complete. Answer each question directly on the examination paper, in the space provided. **Blank pages may be found at the end of the exam and may be used for rough work or in case you need more space for one of your solutions.**

THIS EXAM IS DOUBLE-SIDED.

1: _____/ 7
2: _____/ 6
3: _____/ 6
4: _____/ 6
5: _____/10
6: _____/ 8
7: _____/12
8: _____/10
9: _____/10
TOTAL: _____/75

Question 1. [7 MARKS]

Part (a) [2 MARKS]

Give an example of a number whose decimal representation is finite (i.e., it only has a finite number of nonzero digits) but whose binary representation is not.

Part (b) [2 MARKS]

In a floating-point system with base β, precision p, and rounding to nearest, what is the maximum relative error in representing any nonzero real number within the range of the system? Give an expression for the error.

Part (c) [3 MARKS]

A function $f(x)$ is to be evaluated at each of N equally spaced points in the interval $[0, 1]$, where N is large.

Which of the following two `matlab` scripts is preferable for this task? Explain.

matlab script 1

```
h = 1.0 / (N-1);
x = 0.0;
for i = 1:N
    disp(sprintf('f(%g) = %g', x, f(x)));
    x = x + h;
end
```

matlab script 2

```
h = 1.0 / (N-1);
for i = 1:N
    x = (i-1)*h;
    disp(sprintf('f(%g) = %g', x, f(x)));
end
```

Question 2. [6 MARKS]

Assume that you are solving the quadratic equation $ax^2 + bx + c = 0$, with $a = 1.22$, $b = 3.34$ and $c = 2.28$, using a normalized floating-point system with base $\beta = 10$, precision $p = 3$ and round-to-nearest.

Part (a) [2 MARKS]

What is the computed value of the discriminant $b^2 - 4ac$?

Part (b) [1 MARK]

What is the correct value of the discriminant in real (exact) arithmetic?

Part (c) [2 MARKS]

What is the relative error in the computed value of the discriminant?

Part (d) [1 MARK]

Why is the relative error much larger than the machine epsilon $\frac{1}{2} \times 10^{-2}$?

Question 3. [6 MARKS]

Let x and y be two floating-point numbers. If $x \approx y$, then we would expect some cancellation in computing $\log(x) - \log(y)$. On the other hand, we know that $\log(x) - \log(y) = \log(x/y)$, and that the right-hand-side expression involves no cancellation.

Does this mean that computing $\log(x/y)$ instead of $\log(x) - \log(y)$ is likely to give a better result? Explain. If your answer is no, propose an alternate algorithm for computing a value of $\log(x) - \log(y)$.

Question 4. [6 MARKS]

Part (a) [2 MARKS]

Assume that you have already computed the **LU** factorization, $\mathbf{A} = \mathbf{LU}$, of the nonsingular matrix $\mathbf{A}$. How would you use it to solve the linear system $\mathbf{A}^T \underline{x} = \underline{b}$? ($\mathbf{A}^T$ is the transpose of $\mathbf{A}$.)

Part (b) [2 MARKS]

Write a 4×4 permutation matrix that interchanges the 2nd and 4th components of any 4 element vector.

Write a 4×4 permutation matrix that reverses the order of the components of any 4 element vector.

Part (c) [2 MARKS]

Why is computing the condition number of a general matrix $\mathbf{A}$ a nontrivial problem?

Question 5. [10 MARKS]

Consider the equation $\mathbf{A}\underline{x} = \underline{b}$ with

$$\mathbf{A} = \begin{bmatrix} 2 & -4 & 2 & 4 \\ 1 & -2 & 3 & 3 \\ 0 & 3 & -3 & 3 \\ 0 & -1 & -3 & 1 \end{bmatrix}, \quad \underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \quad \text{and} \quad \underline{b} = \begin{bmatrix} 4 \\ 5 \\ 3 \\ -3 \end{bmatrix}.$$

Part (a) [7 MARKS]

Use Gaussian Elimination **with row pivoting** to compute the $\mathbf{PA} = \mathbf{LU}$ factorization of the coefficient matrix. Please show all steps in your solution and place your final results in the appropriate boxes below.

$$\mathbf{P} = \begin{bmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{bmatrix} \qquad \mathbf{L} = \begin{bmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{bmatrix} \qquad \mathbf{U} = \begin{bmatrix} \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \\ \quad & \quad & \quad & \quad \end{bmatrix}$$

Part (b) [3 MARKS]

Use your $\mathbf{PA} = \mathbf{LU}$ factorization from part (a) to solve the problem $\mathbf{A}\underline{x} = \underline{b}$ for $\underline{x}$. Please show all steps in your solution and place your answer in the box below.

$$\underline{x} = \begin{bmatrix} \quad \\ \quad \\ \quad \\ \quad \\ \quad \end{bmatrix}$$

Question 6. [8 MARKS]

On a computer with no functional unit for floating-point division, one might use multiplication by the reciprocal of the divisor. That is, to compute a/b, multiply a by the reciprocal of b, namely $1/b$. The difficulty then is finding an approximation to the reciprocal of b.

Part (a) [5 MARKS]

Use Newton's method to produce an iterative scheme for approximating the reciprocal of a number $y > 0$. (i.e., to solve the equation $f(x) = (1/x) - y = 0$, given y.) Considering the intended application, your final formula should contain no divisions!

Part (b) [3 MARKS]

For $y > 1$, with what initial guess would you recommend starting the iteration? Are there any initial guesses that you would recommend not using?

Question 7. [12 MARKS]

Part (a) [2 MARKS]

What is the computational cost (number of additions and multiplications) of evaluating a polynomial of degree n using Horner's method?

Part (b) [4 MARKS]

When interpolating a continuous function $f(t)$ by a polynomial, what key factors determine the error in the approximation of the function by the interpolant?

Under what circumstances can the error be large even though the number of interpolation points is large?

Part (c) [2 MARKS]

How does a cubic spline interpolant differ from a piecewise cubic Hermite interpolant?

Part (d) [4 MARKS]

The continuity and smoothness requirements on a cubic spline interpolant still leave two free parameters. Give two examples of additional constraints that might be imposed to determine the cubic spline interpolant to a set of data points. Give the usual name for each of the cubic splines that would be determined.

Question 8. [10 MARKS]

Part (a) [5 MARKS]

Given the three data points

t	-2	0	2
y	1	0	1

,

use a **Newton** basis to construct the polynomial $p_2(t)$ of degree at most 2 that interpolates the data. Show your calculations.

Part (b) [5 MARKS]

The given data was generated by evaluating the function $f(t) = \sin^2\left(\frac{\pi t}{4}\right)$ at the points $t = -2$, 0 and 2. Use the error bound formula for polynomial interpolation to derive an accurate (tight) upper bound for

$$\max_{t\in[-2,2]} |f(t) - p_2(t)|.$$

Give a numerical (decimal) value for your bound. Show your calculations. You may want to use the identity $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$. And don't forget that $\frac{d\sin(u)}{du} = \cos(u)$ and $\frac{d\cos(u)}{du} = -\sin(u)$.

Question 9. [10 MARKS]

Part (a) [4 MARKS]

Consider the data points $\{(0,1),(1,1),(2,5)\}$. Find the piecewise linear interpolant for the data.

Part (b) [6 MARKS]

The `matlab` program

```
x = linspace( 0, pi, n );
y = sin( x );
S = spline( x, y );
```

may be used to determine a not-a-knot cubic spline interpolant of the function $f(x) = \sin(x)$ over the interval $[0, \pi]$. For what values of `n` is the not-a-knot cubic spline guaranteed to have an absolute error that is less than 5.0×10^{-5} over the interval $[0, \pi]$? The variable `n` represents the number of knots (or breakpoints) in the spline.

HINT: The constant is $\dfrac{5}{384}$.

This page for rough work and extra space for solutions.

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Total Marks = 75